A student performed an experiment according to the following steps.

Step 1

Solution 1 was prepared by dissolving a sample of $(NH_4)_2SO_4(s)$ (molar mass 132.14 g/mol) in enough water to make a 2.00 M $(NH_4)_2SO_4(aq)$ solution at 20 °C.

Step 2

Solution 2 was prepared by dissolving a sample of $AgNO_3(s)$ (molar mass 169.87 g/mol) in enough water to make a 2.00 M $AgNO_3(aq)$ solution at 20 °C.

Step 3

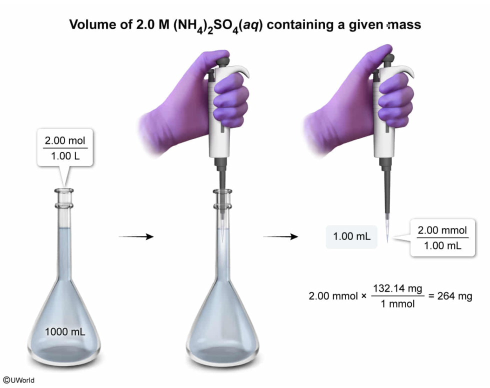
Equal volumes of Solutions 1 and 2 were mixed, resulting in the formation of $NH_4NO_3(aq)$ and a white precipitate of $Ag_2SO_4(s)$ (molar mass 311.79 g/mol).

What volume of Solution 1 would contain 2.64×10^{-1} g of the $(NH_4)_2SO_4$ solute?

- A. 1.00 nL [3%]
- B. 4.00 μL [6%]
- ✔ C. 1.00 mL [78%]
- D. 4.00 mL [11%]

Correct
77% answered correctly

Explanation:



The molar concentration (ie, molarity) of a solution is defined as the moles of solute (ie, a dissolved compound) per liter of solution. However, when dealing with very small volumes or very dilute concentrations of a solution, it is sometimes convenient to scale defined units (eg, molarity) using metric

unit prefixes. For example, if the units in *both* the numerator (moles) *and* the denominator (liters) of molarity are scaled by the *same factor* (eg, 1,000) to give millimoles and milliliters, the factors cancel and yield a result that is numerically the same but has smaller-scale units.

$$2.00\,\frac{\cancel{\text{mol}}}{\cancel{\text{L}}} \times \frac{\cancel{1000}\,\text{mmol}}{1\,\cancel{\text{mol}}} \times \frac{1\,\cancel{\text{L}}}{\cancel{1000}\,\text{mL}} = 2.00\,\frac{\text{mmol}}{\text{mL}}$$

As such, the units of the ratio can be similarly scaled by different factors to yield several equivalent expressions:

$$2.00\ \frac{\text{mol}}{\text{L}} = 2.00\ \frac{\text{mmol}}{\text{mL}} = 2.00\ \frac{\mu\text{mol}}{\mu\text{L}} = 2.00\ \frac{\text{nmol}}{\text{nL}}$$

Likewise, the molar mass of a compound (ie, the amount of **mass contained in 1 mole** of the compound) can be scaled to give equivalent expressions that relate smaller amounts of mass to a correspondingly smaller number of molecules.

$$132.14\ \frac{\text{g}}{\text{mol}} = 132.14\ \frac{\text{mg}}{\text{mmol}} = 132.14\ \frac{\mu\text{g}}{\mu\text{mol}} = 132.14\ \frac{\text{ng}}{\text{nmol}}$$

The mass of 2.64×10^{-1} g specified in the question is equivalent to 264 mg:

$$2.64 \times 10^{-1}\ \text{g} = 0.264\ \text{g}$$

$$0.264\ \cancel{\text{g}} \times \frac{1000\ \text{mg}}{1\ \cancel{\text{g}}} = 264\ \text{mg}$$

Dividing this mass by a correspondingly scaled equivalent of the molar mass of $(NH_4)_2SO_4$ gives the number of millimoles:

$$264\ \cancel{\text{mg}} \times \frac{1\ \text{mmol}}{132.14\ \cancel{\text{mg}}} = 2.00\ \text{mmol}$$

Taking this result and dividing by the molarity of Solution 1 (scaled to matching units) gives the volume of Solution 1 that would contain 2.00 mmol (2.64×10^{-1} g) of the $(NH_4)_2SO_4$ solute.

$$2.00\ \cancel{\text{mmol}} \times \frac{1\ \text{mL}}{2.00\ \cancel{\text{mmol}}} = 1.00\ \text{mL}$$

(Choice A) A volume of 1.00 nL results from mistaking nanograms (ng) for milligrams (mg) in the conversion.

(Choice B) A volume of 4.00 μL results from mistaking micrograms (μg) for milligrams (mg) in the conversion and multiplying instead of dividing by the molarity of Solution 1.

(Choice D) A volume of 4.00 mL results from multiplying instead of dividing by the molarity of Solution 1 in the last step of the calculation.

Educational objective:
In measurements of molarity and molar mass, the numerator and the denominator can be scaled by the same factor to give equivalent expressions that are numerically the same but have larger- or smaller-scale units.

Time spent:0 sec
Subject:
General Chemistry

QID:404015
System:
Atomic Theory and Chemical Composition

A student performed an experiment according to the following steps.

Step 1

Solution 1 was prepared by dissolving a sample of $(NH_4)_2SO_4(s)$ (molar mass 132.14 g/mol) in enough water to make a 2.00 M $(NH_4)_2SO_4(aq)$ solution at 20 °C.

Step 2

Solution 2 was prepared by dissolving a sample of $AgNO_3(s)$ (molar mass 169.87 g/mol) in enough water to make a 2.00 M $AgNO_3(aq)$ solution at 20 °C.

Step 3

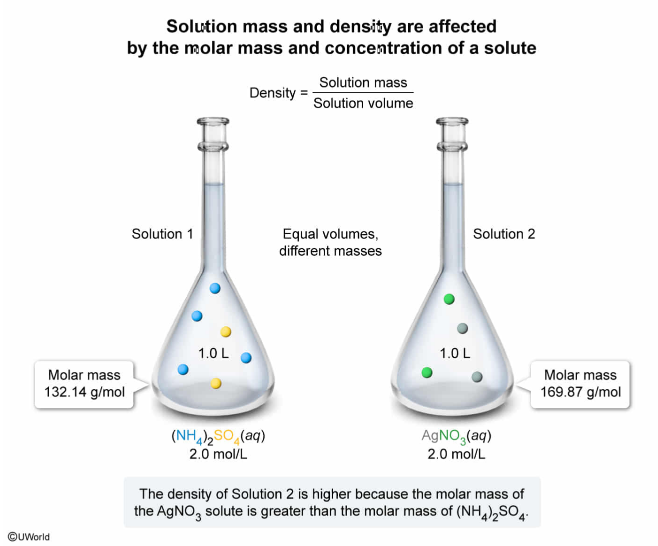
Equal volumes of Solutions 1 and 2 were mixed, resulting in the formation of $NH_4NO_3(aq)$ and a white precipitate of $Ag_2SO_4(s)$ (molar mass 311.79 g/mol).

Compared to the density of Solution 1, the density of Solution 2 is:

- A. lower, because Solution 2 has a larger volume than Solution 1. [4%]
- B. lower, because Solution 1 contains more solute particles than Solution 2. [7%]
- C. the same, because the molar concentration of the solute in each solution is the same. [10%]
- ✔ D. higher, because the molar mass of $AgNO_3$ is greater than the molar mass of $(NH_4)_2SO_4$. [76%]

Correct
75% answered correctly

Explanation:



The **density *ρ*** of a substance is defined as the **mass *m*** it contains per unit of **volume *V*** that it occupies:

$$\rho = \frac{m}{V}$$

When evaluating the density of an aqueous solution, the mass of the solution is the sum of the mass of the solute (ie, a dissolved compound) and the mass of the solvent (ie, water).

$$\rho_{\text{solution}} = \frac{m_{\text{solution}}}{V_{\text{solution}}} = \frac{\left(m_{\text{solute}} + m_{\text{solvent}}\right)}{V_{\text{solution}}}$$

Although the exact masses of the solute and solvent in Solutions 1 and 2 are not given in the passage, a relative comparison of the densities of these solutions can be made based on their molar concentrations (ie, the moles of solute per liter of solution). Because the molar concentrations of the solutions are identical (ie, 2.0 M), 1.00 L of each solution contains the same number of moles of solute formula units.

$$1.00\ \cancel{\text{L Solution 1}} \times \frac{2.00\ \text{mol}\ (\text{NH}_4)_2\text{SO}_4}{1\ \cancel{\text{L Solution 1}}} = 2.00\ \text{mol}\ (\text{NH}_4)_2\text{SO}_4$$

$$1.00\ \cancel{\text{L Solution 2}} \times \frac{2.00\ \text{mol AgNO}_3}{1\ \cancel{\text{L Solution 2}}} = 2.00\ \text{mol AgNO}_3$$

However, the molar masses of the formula units of each solute are very different. Consequently, the $AgNO_3$ solute contributes much more to the mass of Solution 2 than the $(NH_4)_2SO_4$ solute contributes to the mass of Solution 1.

$$2.00\ \cancel{\text{mol}\ (\text{NH}_4)_2\text{SO}_4} \times \frac{132.14\ \text{g}\ (\text{NH}_4)_2\text{SO}_4}{1\ \cancel{\text{mol}\ (\text{NH}_4)_2\text{SO}_4}} = 264.3\ \text{g}\ (\text{NH}_4)_2\text{SO}_4\ \text{in}\ 1.00\ \text{L of Solution 1}$$

$$2.00\ \cancel{\text{mol AgNO}_3} \times \frac{169.87\ \text{g AgNO}_3}{1\ \cancel{\text{mol AgNO}_3}} = 339.7\ \text{g AgNO}_3\ \text{in}\ 1.00\ \text{L of Solution 2}$$

Although the mass of the solvent (water) differs slightly in Solutions 1 and 2 due to water molecules packing differently around the two solutes in the solution volumes, the mass of the solute is the primary factor that distinguishes the solution densities (ie, Solution 2 has much more mass per unit volume due to a more massive solute). Therefore, compared to the density of Solution 1, the density of Solution 2 is higher because the molar mass of $AgNO_3$ is greater than the molar mass of $(NH_4)_2SO_4$.

(Choice A) The exact volumes of the solutions are not known, but the given molar concentrations compare the solutions on the *same volume basis* (ie, moles per *1 L of solution*).

(Choice B) Although Solution 1 does contain more solute particles than Solution 2, the fewer particles in Solution 2 are more massive, which gives Solution 2 a *higher* density than Solution 1.

(Choice C) Although the molar concentrations of the solute in each solution are the same, the molar masses of the solutes (ie, the mass contribution per mole) are different.

Educational objective:
The density of a solution is equal to the mass of the solution (ie, the sum of the mass of the solute and the mass of the solvent) per unit volume of solution. The molar concentration and molar mass of the solute determines the solute contribution to the mass of the solution.

Time spent:0 sec
Subject:
General Chemistry

QID:404016
System:
Atomic Theory and Chemical Composition

A student performed an experiment according to the following steps.

Step 1

Solution 1 was prepared by dissolving a sample of $(NH_4)_2SO_4(s)$ (molar mass 132.14 g/mol) in enough water to make a 2.00 M $(NH_4)_2SO_4(aq)$ solution at 20 °C.

Step 2

Solution 2 was prepared by dissolving a sample of $AgNO_3(s)$ (molar mass 169.87 g/mol) in enough water to make a 2.00 M $AgNO_3(aq)$ solution at 20 °C.

Step 3

Equal volumes of Solutions 1 and 2 were mixed, resulting in the formation of $NH_4NO_3(aq)$ and a white precipitate of $Ag_2SO_4(s)$ (molar mass 311.79 g/mol).

Which statement correctly describes the initial concentrations of the ions in Solutions 1 and 2? The concentration of:

✔ A. $SO_4^{2-}(aq)$ in Solution 1 is 2 times greater than the concentration needed to precipitate all the $Ag^+(aq)$ in Solution 2. [34%]

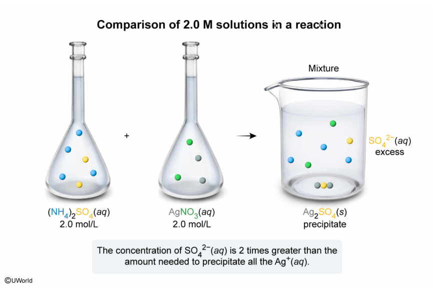
B. $SO_4^{2-}(aq)$ in Solution 1 exactly equals the concentration needed to precipitate all the $Ag^+(aq)$ in Solution 2. [33%]

C. $Ag^+(aq)$ in Solution 2 is 2 times greater than the concentration needed to precipitate all the $SO_4^{2-}(aq)$ Solution 1. [29%]

D. $Ag^+(aq)$ in Solution 2 is 4 times greater than the concentration needed to precipitate all the $SO_4^{2-}(aq)$ in Solution 1. [2%]

Correct
34% answered correctly

Explanation:



The number of moles of one chemical species is related to the number of moles of another chemical species in a reaction by the mole ratios of the stoichiometric coefficients from a balanced reaction equation. As such, the molar concentration (ie, the moles of solute per liter of solution) of a chemical species is useful in evaluating chemical conversions.

Although a balanced chemical reaction is not explicitly given, the passage indicates that $(NH_4)_2SO_4(aq)$ (Solution 1) and $AgNO_3(aq)$ (Solution 2) react upon mixing and form $NH_4NO_3(aq)$ and $Ag_2SO_4(s)$. Stating this description as a balanced reaction equation gives:

$$(NH_4)_2SO_4(aq) + 2\ AgNO_3(aq) \rightarrow 2\ NH_4NO_3(aq) + Ag_2SO_4(s)$$

Removing the spectator ions simplifies the reaction to give the balanced net ionic equation, which shows only the reacting ions:

$$SO_4^{2-}(aq) + 2\ Ag^+(aq) \rightarrow Ag_2SO_4(s)$$

In the reaction of an equal volume (eg, 1.00 L) of Solutions 1 and 2, the initial number of moles of the reacting ion present in each solution is:

$$1.00\ \cancel{\text{L Solution 1}} \times \frac{2.00\ \cancel{\text{mol }(\text{NH}_4)_2\text{SO}_4(\text{aq})}}{1\ \cancel{\text{L Solution 1}}} \times \frac{1\text{ mol SO}_4^{2-}(\text{aq})}{1\ \cancel{\text{mol }(\text{NH}_4)_2\text{SO}_4(\text{aq})}} = 2.00\text{ mol SO}_4^{2-}(\text{aq})$$

$$1.00\ \cancel{\text{L Solution 2}} \times \frac{2.00\ \cancel{\text{mol AgNO}_3(\text{aq})}}{1\ \cancel{\text{L Solution 2}}} \times \frac{1\text{ mol Ag}^+(\text{aq})}{1\ \cancel{\text{mol AgNO}_3(\text{aq})}} = 2.00\text{ mol Ag}^+(\text{aq})$$

According to the mole ratios of the balanced net ionic equation, the number of moles of SO_4^{2-}(*aq*) needed to precipitate all the Ag^+(*aq*) in the mixture is:

$$2.00\ \cancel{\text{mol Ag}^+(\text{aq})} \times \frac{1\text{ mol SO}_4^{2-}(\text{aq})}{2\ \cancel{\text{mol Ag}^+(\text{aq})}} = 1.00\text{ mol SO}_4^{2-}(\text{aq})\text{ needed}$$

However, there are 2.00 mol of SO_4^{2-}(*aq*) present in the mixture. Dividing the moles of SO_4^{2-}(*aq*) present by the moles of SO_4^{2-}(*aq*) needed gives:

$$\frac{2.00\ \cancel{\text{mol SO}_4^{2-}(\text{aq})\text{ present}}}{1.00\ \cancel{\text{mol SO}_4^{2-}(\text{aq})\text{ needed}}} = 2\text{ times (excess)}$$

Therefore, the concentration of SO_4^{2-}(*aq*) in Solution 1 is 2 times greater than the concentration needed to precipitate all the Ag^+(*aq*) in Solution 2.

(Choice B) The initial concentrations of Ag^+(*aq*) and SO_4^{2-}(*aq*) are equal, but the mole ratio of the balanced net ionic equation shows that precipitating a given number of moles of Ag^+(*aq*) takes *half* as many moles of SO_4^{2-}(*aq*).

(Choices C and D) According to the mole ratios of the balanced net ionic equation, the number of moles of Ag^+(*aq*) needed to precipitate all the SO_4^{2-}(*aq*) in the mixture is:

$$2.00\ \cancel{\text{mol SO}_4^{2-}(\text{aq})} \times \frac{2\text{ mol Ag}^+(\text{aq})}{1\ \cancel{\text{mol SO}_4^{2-}(\text{aq})}} = 4.00\text{ mol Ag}^+(\text{aq})\text{ needed}$$

However, there are only 2.00 mol of Ag^+(*aq*) present in the mixture. The ratio of these amounts is:

$$\frac{2.00\ \cancel{\text{mol Ag}^+(\text{aq})\text{ present}}}{4.00\ \cancel{\text{mol Ag}^+(\text{aq})\text{ needed}}} = \frac{1}{2}$$

Therefore, the concentration of Ag^+(*aq*) is only *half* of the concentration needed to precipitate all the SO_4^{2-}(*aq*).

Educational objective:
The molar concentration of a reactant indicates the number of moles per liter of solution, and stoichiometric mole ratios from a balanced reaction equation give the proportions by which the moles of one chemical species reacts with the moles of another.

Time spent: 0 sec
Subject:
General Chemistry

QID: 404017
System:
Atomic Theory and Chemical Composition

A student performed an experiment according to the following steps.

Step 1

Solution 1 was prepared by dissolving a sample of $(NH_4)_2SO_4(s)$ (molar mass 132.14 g/mol) in enough water to make a 2.00 M $(NH_4)_2SO_4(aq)$ solution at 20 °C.

Step 2

Solution 2 was prepared by dissolving a sample of $AgNO_3(s)$ (molar mass 169.87 g/mol) in enough water to make a 2.00 M $AgNO_3(aq)$ solution at 20 °C.

Step 3

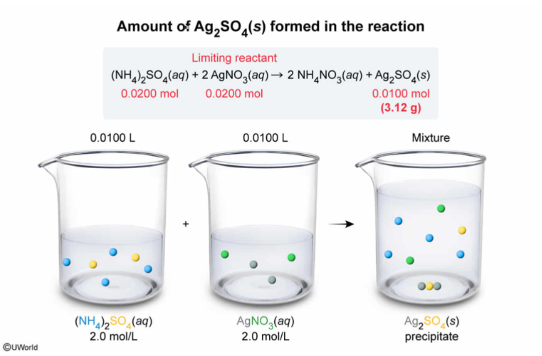
Equal volumes of Solutions 1 and 2 were mixed, resulting in the formation of $NH_4NO_3(aq)$ and a white precipitate of $Ag_2SO_4(s)$ (molar mass 311.79 g/mol).

When 10.0 mL of Solution 1 and 10.0 mL of Solution 2 are mixed, what is the mass of $Ag_2SO_4(s)$ produced in the reaction? (Note: Assume that the conversion of the limiting reactant goes to completion and that all the product precipitates from the solution.)

- A. 1.70 g [8%]
- B. 2.64 g [16%]
- ✔ C. 3.12 g [51%]
- D. 6.24 g [24%]

Correct
49% answered correctly

Explanation:



In a chemical reaction, the products form from the reactants in the proportions indicated the balanced reaction equation. The amount of product that can be formed in a chemical reaction is determined by how much of the limiting reactant (ie, the reactant that gets used up first) is present in the reaction mixture. As such, the molar concentration (molarity) (ie, the moles of solute per liter of solution) of a chemical species is useful in chemical reaction calculations.

According to the passage, mixing $(NH_4)_2SO_4$(*aq*) (Solution 1) and $AgNO_3$(*aq*) (Solution 2) results in a reaction that forms NH_4NO_3(*aq*) and Ag_2SO_4(*s*). As such, the balanced equation for the reaction is:

$$(NH_4)_2SO_4(aq) + 2\ AgNO_3(aq) \rightarrow 2\ NH_4NO_3(aq) + Ag_2SO_4(s)$$

Because the molar concentrations of $(NH_4)_2SO_4$(*aq*) and $AgNO_3$(*aq*) are equal (ie, 2.0 M) but the reaction occurs in a 1:2 ratio, the concentration of SO_4^{2-}(*aq*) is two times greater than the concentration needed to precipitate all the Ag^+(*aq*). Consequently, the limiting reactant must be $AgNO_3$(*aq*). This can be confirmed by calculating the amount of Ag_2SO_4(*s*) theoretically produced from the stated amounts of reactants (ie, the limiting reactant will yield the smallest result). Accordingly, the molarity of $AgNO_3$(*aq*) multiplied by the reacting volume (in liters) gives the amount of $AgNO_3$(*aq*) initially present in solution:

$$0.0100\ \cancel{\text{L Solution 2}} \times \frac{2.00\ \text{mol AgNO}_3(\text{aq})}{1\ \cancel{\text{L Solution 2}}} = 0.0200\ \text{mol AgNO}_3(\text{aq})$$

Applying the mole ratio of Ag_2SO_4(*s*) to $AgNO_3$(*aq*) from the balanced reaction gives the moles of Ag_2SO_4(*s*) formed:

$$0.0200\ \cancel{\text{mol AgNO}_3(\text{aq})} \times \frac{1\ \text{mol Ag}_2\text{SO}_4(s)}{2\ \cancel{\text{mol AgNO}_3(\text{aq})}} = 0.0100\ \text{mol Ag}_2\text{SO}_4(s)$$

Multiplying the moles of Ag_2SO_4(*s*) by its molar mass yields the mass precipitated:

$$0.0100\ \cancel{\text{mol Ag}_2\text{SO}_4(s)} \times \frac{311.79\ \text{g Ag}_2\text{SO}_4(s)}{1\ \cancel{\text{mol Ag}_2\text{SO}_4(s)}} = 3.12\ \text{g Ag}_2\text{SO}_4(s)$$

(Choice A) A value of 1.70 g results from using the molar mass of $AgNO_3$ instead of the molar mass of Ag_2SO_4 in the last step of the calculation.

(Choice B) A quantity of 2.64 g is the mass of $(NH_4)_2SO_4$ initially present in the reaction mixture.

(Choice D) A value of 6.24 g results either from incorrectly selecting $(NH_4)_2SO_4$(*aq*) as the limiting reactant or from assuming a 1:1 mole ratio of Ag_2SO_4(*s*) to $AgNO_3$(*aq*) during the second step of the calculation.

Educational objective:
The products of a chemical reaction form from the reactants in the proportions indicated by the mole ratios of the stoichiometric coefficients of the balanced reaction equation. The amount of product that can be formed in a reaction is determined by how much of the limiting reactant is present in the reaction mixture.

Time spent:0 sec
Subject:
General Chemistry

QID:404018
System:
Atomic Theory and Chemical Composition